Module 10 : Crossover randomized experiments

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Design of Experiments - Stat 140

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Example 1

doi: 10.1111/fcp.123

Cannabis smoking impairs driving performance on the simulator and real driving: a randomized, double-blind, placebo-controlled, crossover trial

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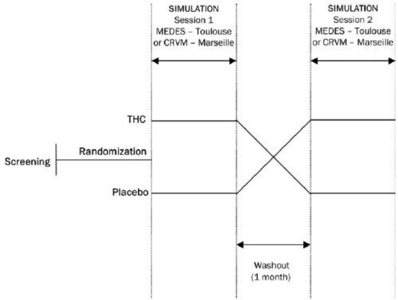

Example 1

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Cannabis smoking impairs driving performance on the simulator and real driving: a randomized, double-blind, placebo-controlled, crossover trial

Model tested:

$$\begin{aligned}\text{Outcomes} = &\beta 0 + \beta 1.\text{treatment} + \beta 2.\text{sequence} \\ &+ \beta 3.\text{driving} + \beta 4.\text{session} \\ &+ \beta 5.(\text{driving} \times \text{treatment}) \\ &+ \text{subject}(\text{sequence}) + \varepsilon.\end{aligned}$$

ILC was log-transformed for analysis. Correlations between outcomes were calculated using Spearman's coefficient. *P*-values <0.05 were considered significant. The consistency and analyzes were performed with SAS software version 9.2 (SAS Institute).

Example 2

Aspartame ingestion and headaches:
A randomized crossover trial

S.K. Van Den Eeden, PhD; T.D. Koepsell, MD, MPH; W.T. Longstreth, Jr, MD, MPH;
G. van Belle, PhD; J.R. Daling, PhD; and B. McKnight, PhD

Example 2

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Table 1. Treatment sequences*

	Period			
Sequence	**One**	**Two**	**Three**	**Four**
A	Aspartame	Placebo	Aspartame	Placebo
B	Aspartame	Placebo	Placebo	Aspartame
C	Placebo	Aspartame	Placebo	Aspartame
D	Placebo	Aspartame	Aspartame	Placebo

* All subjects started with a 7-day placebo run-in period.

Aspartame ingestion and headaches:
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Data analysis. The mean number of days with headache was computed for each subject and treatment period by dividing the number of days headaches were reported by the number of days on the treatment. Except where noted, data from the run-in period were included in the calculations for placebo treatment, since subjects were unaware of the crossover design and did not know that the first week consisted of a placebo run-in period. The means and standard deviations presented in the tables were calculated from individual means. The subjects in the "not very sure" (n = 3) and the "do not know" (n = 2) groups were combined for the analysis by strength of prior belief.

Because some subjects did not complete the full protocol, it was necessary to compare the probability of headache during aspartame and placebo treatment, accounting, where needed, for differing lengths of treatment. Large sample methods based on the binomial distribution for number of days with headache out of the variable number of days on study during each treatment period were used to estimate the within-subject relative risk of a headache, comparing aspartame with placebo, and to test whether it equaled one.[27] This analytic method does not impute any data beyond what was observed in the study; instead, it makes maximum use of available data, including data on subjects who completed only a part of the protocol.

Crossover randomized experiments

- Target : All potential outcomes, similar as in twin studies and individual effects.
- Goes one step further than matched pair design.
- Commonly-used in randomized experiments with pain because individual heterogeneity (no objective scale).
- Problem ?
- Important for observational studies
- Consider a population of N participants, each of which can be exposed to any one of two blinded sequences of treatments : treatment followed by control or control followed by treatment.

Assignment mechanism

- Let W be an $Nx2$ matrix with i^{th} row $W_i = (W_{i,j=1}, W_{i,j=2})$, where we index by $j \in \{1, 2\}$ the two periods for participant i.

- In a crossover experiment, one possible assignment mechanism :
    - $W_{i,j=1}$ is Bernoulli with probability $\frac{1}{2}$
    - $W_{i,j=2} = 1 - W_{i,j=1}$.

- There are 2^N possible ways in which N experimental participants can be randomly allocated to the two sequences of treatments.

- Possible assignment mechanism :

$$P(\mathbf{W}|\mathbf{X},\mathbf{Y(0)},\mathbf{Y(1)}) = \begin{cases} \frac{1}{2^N} & \text{if } W \in \mathbb{W}^+ \\ 0 & \text{otherwise.} \end{cases}$$

where $\mathbb{W}^+$ denotes the set of all possible values of W.

Other possible assignment mechanism for randomized crossover experiment

- Completely randomized assignment mechanism :
  - $\mathrm{P}(W_{i,j=1}=1)=\frac{N_{T,j=1}}{N}$
  - $W_{i,j=2} = 1 - W_{i,j=1}$.

- Assignment mechanism :

$$P(\mathbf{W}|\mathbf{X},\mathbf{Y(0)},\mathbf{Y(1)}) = \begin{cases} \binom{N}{N_{t,j=1}}^{-1} & \text{if } \sum_{i=1}^{N} W_{i,j=1} = N_{t,j=1} \\ & \text{and } W_{i,j=2} = 1 - W_{i,j=1} \\ & \\ 0 & \text{otherwise.} \end{cases}$$

Science Table

- Let $Y_{i,j}(W_{i,j} = w_{i,j})$ denote the ith participant's potential outcome at period j.

i	$Y_{i,j=1}(W_{i,j=1} = 0)$	$Y_{i,j=1}(W_{i,j=1} = 1)$	$Y_{i,j=2}(W_{i,j=2} = 0)$	$Y_{i,j=2}(W_{i,j=2} = 1)$
1	$Y_{1,j=1}(W_{1,j=1} = 0)$	$Y_{1,j=1}(W_{1,j=1} = 1)$	$Y_{1,j=2}(W_{1,j=2} = 0)$	$Y_{1,j=2}(W_{1,j=2} = 1)$
2	$Y_{2,j=1}(W_{2,j=1} = 0)$	$Y_{2,j=1}(W_{2,j=1} = 1)$	$Y_{2,j=2}(W_{2,j=2} = 0)$	$Y_{2,j=2}(W_{2,j=2} = 1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_{N,j=1}(W_{N,j=1} = 0)$	$Y_{N,j=1}(W_{N,j=1} = 1)$	$Y_{N,j=2}(W_{N,j=2} = 0)$	$Y_{N,j=2}(W_{N,j=2} = 1)$

- Causal estimands follow from the science table

Science Table

- Let $Y_{i,j}(W_{i,j} = w_{i,j})$ denote the ith participant's potential outcome at period j.

i	$Y_{i,j=1}(W_{i,j=1} = 0)$	$Y_{i,j=1}(W_{i,j=1} = 1)$	$Y_{i,j=2}(W_{i,j=2} = 0)$	$Y_{i,j=2}(W_{i,j=2} = 1)$
1	$Y_{1,j=1}(W_{1,j=1} = 0)$	$Y_{1,j=1}(W_{1,j=1} = 1)$	$Y_{1,j=2}(W_{1,j=2} = 0)$	$Y_{1,j=2}(W_{1,j=2} = 1)$
2	$Y_{2,j=1}(W_{2,j=1} = 0)$	$Y_{2,j=1}(W_{2,j=1} = 1)$	$Y_{2,j=2}(W_{2,j=2} = 0)$	$Y_{2,j=2}(W_{2,j=2} = 1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_{N,j=1}(W_{N,j=1} = 0)$	$Y_{N,j=1}(W_{N,j=1} = 1)$	$Y_{N,j=2}(W_{N,j=2} = 0)$	$Y_{N,j=2}(W_{N,j=2} = 1)$

- Causal estimands follow from the science table
  - Unit-level (i.e., participant i at period j) causal effect of treatment versus control :

$$\tau_{i,j} = Y_{i,j}(W_{i,j} = 1) - Y_{i,j}(W_{i,j} = 0)$$

Science Table

- Let $Y_{i,j}(W_{i,j} = w_{i,j})$ denote the ith participant's potential outcome at period j.

i	$Y_{i,j=1}(W_{i,j=1} = 0)$	$Y_{i,j=1}(W_{i,j=1} = 1)$	$Y_{i,j=2}(W_{i,j=2} = 0)$	$Y_{i,j=2}(W_{i,j=2} = 1)$
1	$Y_{1,j=1}(W_{1,j=1} = 0)$	$Y_{1,j=1}(W_{1,j=1} = 1)$	$Y_{1,j=2}(W_{1,j=2} = 0)$	$Y_{1,j=2}(W_{1,j=2} = 1)$
2	$Y_{2,j=1}(W_{2,j=1} = 0)$	$Y_{2,j=1}(W_{2,j=1} = 1)$	$Y_{2,j=2}(W_{2,j=2} = 0)$	$Y_{2,j=2}(W_{2,j=2} = 1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_{N,j=1}(W_{N,j=1} = 0)$	$Y_{N,j=1}(W_{N,j=1} = 1)$	$Y_{N,j=2}(W_{N,j=2} = 0)$	$Y_{N,j=2}(W_{N,j=2} = 1)$

- Causal estimands follow from the science table
  - Period-average participant-level causal effect of treatment versus control :

$$\tau_i = \frac{1}{2}(\tau_{i,1} + \tau_{i,2})$$

Science Table

- Let $Y_{i,j}(W_{i,j} = w_{i,j})$ denote the ith participant's potential outcome at period j.

i	$Y_{i,j=1}(W_{i,j=1} = 0)$	$Y_{i,j=1}(W_{i,j=1} = 1)$	$Y_{i,j=2}(W_{i,j=2} = 0)$	$Y_{i,j=2}(W_{i,j=2} = 1)$
1	$Y_{1,j=1}(W_{1,j=1} = 0)$	$Y_{1,j=1}(W_{1,j=1} = 1)$	$Y_{1,j=2}(W_{1,j=2} = 0)$	$Y_{1,j=2}(W_{1,j=2} = 1)$
2	$Y_{2,j=1}(W_{2,j=1} = 0)$	$Y_{2,j=1}(W_{2,j=1} = 1)$	$Y_{2,j=2}(W_{2,j=2} = 0)$	$Y_{2,j=2}(W_{2,j=2} = 1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_{N,j=1}(W_{N,j=1} = 0)$	$Y_{N,j=1}(W_{N,j=1} = 1)$	$Y_{N,j=2}(W_{N,j=2} = 0)$	$Y_{N,j=2}(W_{N,j=2} = 1)$

- Causal estimands follow from the science table
  - Population-average period-level causal effect of treatment versus control :

$$\tau_j = \frac{1}{N} \sum_{i=1}^{N} \tau_{i,j}$$

Science Table

- Let $Y_{i,j}(W_{i,j} = w_{i,j})$ denote the ith participant's potential outcome at period j.

i	$Y_{i,j=1}(W_{i,j=1} = 0)$	$Y_{i,j=1}(W_{i,j=1} = 1)$	$Y_{i,j=2}(W_{i,j=2} = 0)$	$Y_{i,j=2}(W_{i,j=2} = 1)$
1	$Y_{1,j=1}(W_{1,j=1} = 0)$	$Y_{1,j=1}(W_{1,j=1} = 1)$	$Y_{1,j=2}(W_{1,j=2} = 0)$	$Y_{1,j=2}(W_{1,j=2} = 1)$
2	$Y_{2,j=1}(W_{2,j=1} = 0)$	$Y_{2,j=1}(W_{2,j=1} = 1)$	$Y_{2,j=2}(W_{2,j=2} = 0)$	$Y_{2,j=2}(W_{2,j=2} = 1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_{N,j=1}(W_{N,j=1} = 0)$	$Y_{N,j=1}(W_{N,j=1} = 1)$	$Y_{N,j=2}(W_{N,j=2} = 0)$	$Y_{N,j=2}(W_{N,j=2} = 1)$

- Causal estimands follow from the science table
  - Period- and population-average causal effect of treatment versus control :

$$\tau = \frac{1}{2}(\tau_{j=1} + \tau_{j=2})$$

Science Table

- Let $Y_{i,j}(W_{i,j} = w_{i,j})$ denote the ith participant's potential outcome at period j.

i	$Y_{i,j=1}(W_{i,j=1} = 0)$	$Y_{i,j=1}(W_{i,j=1} = 1)$	$Y_{i,j=2}(W_{i,j=2} = 0)$	$Y_{i,j=2}(W_{i,j=2} = 1)$
1	$Y_{1,j=1}(W_{1,j=1} = 0)$	$Y_{1,j=1}(W_{1,j=1} = 1)$	$Y_{1,j=2}(W_{1,j=2} = 0)$	$Y_{1,j=2}(W_{1,j=2} = 1)$
2	$Y_{2,j=1}(W_{2,j=1} = 0)$	$Y_{2,j=1}(W_{2,j=1} = 1)$	$Y_{2,j=2}(W_{2,j=2} = 0)$	$Y_{2,j=2}(W_{2,j=2} = 1)$
...	...	...	...	...
...	...	...	...	...
N	$Y_{N,j=1}(W_{N,j=1} = 0)$	$Y_{N,j=1}(W_{N,j=1} = 1)$	$Y_{N,j=2}(W_{N,j=2} = 0)$	$Y_{N,j=2}(W_{N,j=2} = 1)$

- Causal estimands follow from the science table

- Note that, for each participant i, only two of the four potential outcomes can be observed.

Observed Table

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

- How to write $Y_{i,j}^{obs}$ in terms of $W_{i,j}$ and the potential outcomes ?

Observed Table

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

- How to write $Y_{i,j}^{obs}$ in terms of $W_{i,j}$ and the potential outcomes?

$$Y_{i,j}^{obs} = W_{i,j} Y_{i,j}(W_{i,j} = 1) + (1 - W_{i,j}) Y_{i,j}(W_{i,j} = 0)$$

Fisherian inference

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

- Sharp null hypothesis

Fisherian inference

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

- Test statistic

Fisherian inference

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

- Test statistic

$$T_{paired} = \frac{\frac{1}{N}\sum_{i=1}^{N} d_i}{s_D/\sqrt{N}}$$

- $d_i = \begin{cases} Y_{i,j=1}(W_{i,j=1}=1) - Y_{i,j=2}(W_{i,j=2}=0) & \text{if } W_i = (1,0) \\ Y_{i,j=2}(W_{i,j=2}=1) - Y_{i,j=1}(W_{i,j=1}=0) & \text{if } W_i = (0,1) \end{cases}$
- $s_D^2 = \frac{1}{N-1}\sum_{i=1}^{N}(d_i - \frac{1}{N}\sum_{i=1}^{N} d_i)^2$

Neymanian inference

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

- Estimator for τ_i :

Neymanian inference

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

- Estimator for τ_i :

- Estimator for τ_j :

Neymanian inference

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

- Estimator for τ_i :
- Estimator for τ_j :
- Estimator for τ :

Carry-over and time effects

i	W_i	$Y_{i,j=1}(W_{i,j=1}=0)$	$Y_{i,j=1}(W_{i,j=1}=1)$	$Y_{i,j=2}(W_{i,j=2}=0)$	$Y_{i,j=2}(W_{i,j=2}=1)$
1	(0,1)	$Y_{1,j=1}(W_{1,j=1}=0)$	?	?	$Y_{1,j=2}(W_{1,j=2}=1)$
2	(1,0)	?	$Y_{2,j=1}(W_{2,j=1}=1)$	$Y_{2,j=2}(W_{2,j=2}=0)$	?
...	...	...	...	...	...
...	...	...	...	...	...
N	(0,1)	$Y_{N,j=1}(W_{N,j=1}=0)$	?	?	$Y_{N,j=2}(W_{N,j=2}=1)$

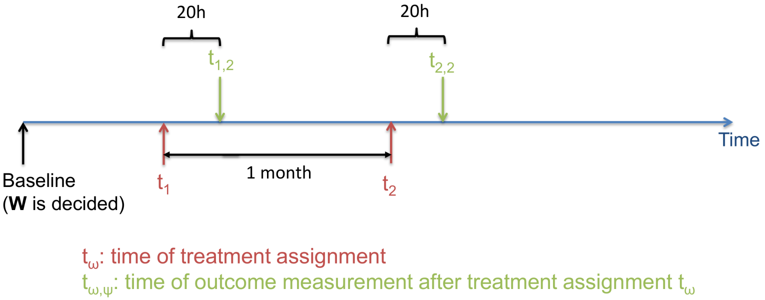
- How to assess carry-over and time effects ?

Application to epigenetics

Description of the crossover experiment.

# of participants	W_i=($W_{i,j=1}$,$W_{i,j=2}$)	Visit 1	Visit 2
0	(0,0)	`Clean air`	`Clean air`
7	(0,1)	`Clean air`	`Ozone`
10	(1,0)	`Ozone`	`Clean air`
0	(1,1)	`Ozone`	`Ozone`

Application to epigenetics



- W_i=(Ozone, Clean air) or W_i=(Clean air, Ozone)
- Y=DNA methylation measured 20h after.